Textbook Decoding Workspace: Particle in a 1 D Box

Course: Structure-Property Relationships in Chemistry

Name: _____ **Date:** _____ **Section:** _____

Instructions

1. Open the Quanta by Atkins textbook to Chapter 2, Nanosystems in 1 Dimension and find the relevant section 2.2 covering the **Particle in a 1D Box**.
2. Follow the mathematical progression line-by-line from the initial setup to the final energy quantization expression.
3. Complete every blank cell in the matrix below by showing work or finding specific definitions/quotes from the text.

Part 1: The Derivation Translation Matrix

Derivation Step	**1. Mathematical Action** *(What operation is happening?)*	**2. Physical Reality** *(What does this mean for the particle?)*	**3. The Text Verdict** *(Textbook rules / quotes)*
Step 1: The Setup $V(x) = 0$ for $0 < x < L$ $V(x) = \infty$ elsewhere	What exact constraints are being placed on the variable x? What does L represent geometrically?	Why must the potential energy $V(x)$ be infinitely high outside the box? Where is the particle completely forbidden from existing?	Quote the definition or description your textbook gives for a **Boundary Condition**:
Step 2: The Equation $-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi$	Identify the calculus operation being performed on the wavefunction ψ. What does $\hbar$ mean in terms of h?	What physical property of the particle does the operator on the left side $(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2})$ calculate?	Write out the textbook's definition of the **Hamiltonian Operator** $(\hat{H})$:
Step 3: General Solution $\psi(x) = A\sin(kx) + B\cos(kx)$	Why is a combination of sine and cosine functions a valid mathematical solution to the second-derivative equation in Step 2?	What are A and B? Are they fixed parameters or arbitrary constants at this point in the derivation?	Copy the text's equation or sentence that defines the value of the constant k in terms of mass (m) and energy (E): $k =$ _____
Step 4: Boundary 1 Evaluating $\psi(0) = 0$ $\implies B = 0$	Substitute $x = 0$ into the equation from Step 3. Show the algebraic steps that force B to equal 0:	Why must the value of the wavefunction ψ drop precisely to zero right at the boundary walls?	Find the section on wave properties. What term does the text use to describe a point where a wave has zero amplitude? Answer: _____
Step 5: Boundary 2 Evaluating $\psi(L) = 0$ $\implies k = \frac{n\pi}{L}$	If $\psi(L) = A\sin(kL) = 0$, and $A \neq 0$, what values must the argument (kL) take? Show how $k = \frac{n\pi}{L}$.	What is the restriction on the integer n? Can $n = 0$? Explain what happens physically to the wave if $n = 0$.	Quote the text explaining how matching a wave to fixed boundaries naturally produces **Quantization**:

Derivation Step	1. Mathematical Action *(What operation is happening?)*	2. Physical Reality *(What does this mean for the particle?)*	3. The Text Verdict *(Textbook rules / quotes)*
Step 6: Normalization $\int_0^L \|\psi(x)\|^2 dx = 1$ $\implies A = \sqrt{\frac{2}{L}}$	What integration rule or identity is used to solve $\int_0^L A^2 \sin^2\left(\frac{n\pi x}{L}\right) dx = 1$?	What is the physical interpretation of $\|\psi(x)\|^2$? Why must the total integrated area across the box equal exactly 1?	State Born's interpretation of the wavefunction as described in your textbook:
Step 7: Quantized Energy $E_n = \frac{n^2 h^2}{8mL^2}$	Equate the expression for k from Step 3 with the expression for k from Step 5. Solve for E to show this final formula:	According to this formula, what happens to the energy levels if the width of the box (L) gets infinitely large?	Find and write out the textbook definition for "**Zero-Point Energy**". Why can't a confined quantum particle be completely at rest?

Part 2: Applying the Derivation

1. Visualizing the Probability Densities

In the space below, sketch the normalized wavefunctions (ψ_n) and the probability densities ($|\psi_n|^2$) for the first three allowed energy states ($n = 1, 2, 3$). Clearly label the positions of any nodes along the x-axis.

2. The Macroscopic Transition

Suppose a baseball ($m = 0.145$ kg) is confined to a macroscopic box that is 1.0 meter wide. Using the energy equation derived in Step 7, explain why we do not observe quantum energy quantization or discrete "allowed tracks" when throwing a baseball in the real world. Support your answer with a qualitative scale argument based on Planck's constant (h).

